Catastrophe Pricing

Commercial Property Insurance

Red Umbrella Insurance — Actuarial Pricing Analysis

Travelers Actuarial Case Competition 2026

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Executive Summary

This report presents a comprehensive catastrophe (CAT) pricing analysis for Red Umbrella Insurance's commercial property portfolio. The analysis covers three primary CAT perils — Hurricane, Hail, and Wildfire — across four geographic regions: Northeast, South, Midwest, and West. The total countrywide Total Insured Limit (TIL) is approximately $253.7 billion.

The analysis integrates two complementary methodologies: 20 years of adjusted historical loss experience (2006–2025) and third-party stochastic CAT model output, including Average Annual Loss (AAL) and Exceedance Probability (EP) curves. Pure premium selections were made at the peril-region level using peril-specific blending ratios of historical and model indications, with risk loads calibrated to tail risk as measured by the ratio of 100-year Probable Maximum Loss (PML) to AAL.

Key findings include:

- Hurricanes represent the highest tail risk, with a countrywide 100-year PML of $609M vs. an AAL of $87M (ratio: ~7.0×)
- Wildfire risk in the West is the fastest-growing exposure, with a 100-year PML of $201M and a PML-to-AAL ratio of ~6.0×
- Hail is the highest-frequency peril but carries significantly lower tail risk (countrywide PML-to-AAL ratio: ~1.8×)
- Proposed rate changes range from –20% (Hail/West) to +46% (Hail/South), driven by divergences between current loads and selected CAT pure premiums
- Regulatory feedback was addressed through EP curve-based justifications for the Northeast and a multi-pronged risk management strategy for the West

1. Background & Scope

Role and Objectives

As CAT Pricing Lead at Red Umbrella Insurance, the objectives of this analysis are to:

- Evaluate 20 years of adjusted historical CAT loss data at the peril and region level
- Assess third-party CAT model output, including AAL and EP curves produced on Red Umbrella's current book of business
- Select CAT pure premium loads for each peril-region combination based on a synthesis of historical and modeled indications
- Advise management on proposed strategies, including risk load structures and potential reinsurance arrangements
- Draft responses to regulatory feedback from the West and Northeast regions

Scope of Coverage

Red Umbrella specializes in commercial property insurance and faces CAT exposure from three perils across four regions:

Peril	Characteristics
Hurricane	Wind damage to roofs and siding; potential total losses; geographic concentration in South and Northeast coastal areas; extremely high tail risk
Hail	Widespread roof damage; rarely total loss; high-frequency, moderate-severity; most prevalent in Midwest and South; expanding suburban footprint increases exposure
Wildfire	Typically total loss events; highest concentration in West; increasing frequency due to drought, urban expansion into wildland-urban interface

Regions analyzed: Northeast, South, Midwest, and West. The exposure base used throughout is Total Insured Limit (TIL), consistent with commercial property industry practice.

Region	TIL ($)	Share of Total (%)
Northeast	$45,129,000,000	17.8%
South	$74,839,000,000	29.5%
Midwest	$77,020,000,000	30.4%
West	$56,710,000,000	22.4%
Countrywide	**$253,698,000,000**	**100.0%**

Key Assumptions

- All historical losses have been developed, trended, and adjusted to reflect current exposure levels and loss cost trends. Historical losses are treated as plausible indications for Red Umbrella's current book of business
- CAT model output (AAL, EP curves) was produced using Red Umbrella's current book of business
- Expenses and LAE are out of scope; the analysis focuses on pure premium and risk load components

- TIL is expressed in dollars; pure premiums are stated per $1 million of TIL for ease of comparison

2. Historical CAT Loss Analysis

Overview and Methodology

Historical loss experience spanning 2006–2025 (20 years) was analyzed at the peril-region level. Three core metrics were calculated: claim frequency (claims per $1 million TIL), claim severity (loss per claim), and pure premium (loss per $1 million TIL). Pure premium equals the product of frequency and severity, and serves as the primary input for historical indications.

All losses are pre-adjusted for trend and exposure changes. The historical database covers 240 peril-year-region data points across 3 perils, 4 regions, and 20 years.

Countrywide Loss Metrics (20-Year Average)

Peril	Freq. (per $M TIL)	Avg. Severity ($)	Pure Premium (per $M TIL)
Hail	0.003388	$146,583	$496
Hurricane	0.004962	$62,670	$311
Wildfire	0.000536	$259,955	$157
Total (All Perils)	**0.008886**	**N/A (blended)**	**$1,043**

Key Observations by Peril

Hurricane

Hurricane losses are characterized by extreme volatility. Most years record zero losses, but individual event years drive enormous aggregate losses — most notably 2019, in which hurricane losses in the South ($453M) produced a countrywide pure premium of over $2,100 per $M TIL, more than double the 20-year average. This event alone illustrates why historical averages are inadequate for hurricane pricing without model supplementation. The South and Northeast are the only affected regions (hurricane risk does not materially reach the Midwest or West).

Observed peril distribution: The South accounts for ~74% of hurricane losses; the Northeast accounts for the remaining ~26%. The Northeast's historical hurricane losses are modest, but the presence of a dense, highly-insured coastal population creates outsized tail risk that historical data alone does not capture.

Hail

Hail is the most frequent CAT peril, with a 20-year countrywide average frequency of 0.003388 per $M TIL. However, due to lower individual claim severity relative to hurricanes and wildfires, hail

rarely produces total losses. The Midwest dominates hail exposure (~70% of countrywide hail losses), consistent with the geographic concentration of severe convective storm systems. The South represents ~22% of countrywide hail exposure, a share that is growing as suburban expansion in Southern states creates a larger insured footprint. The West and Northeast contribute minimally.

A notable trend is the increase in hail pure premiums in the shorter-term averages relative to the 20-year average in the Northeast and South, suggesting that hail risk may be intensifying in these regions due to expansion of insured property into high-exposure corridors.

Wildfire

Wildfire is the lowest-frequency peril but the highest-severity on a per-claim basis (20-year countrywide average severity: ~$260,000). The West accounts for ~87% of countrywide wildfire losses by TIL distribution. Loss experience shows meaningful year-over-year variability, with large loss years in 2016, 2019, 2021, and 2023/2025 in the West region. A 10-year pure premium of approximately $1,172 per $M TIL in the West versus a 20-year average of $666 suggests an escalating trend, which aligns with known macro drivers: drought frequency, WUI expansion, and dry fuel accumulation.

Pure Premium Trends by Time Horizon

To inform selections, historical pure premiums were computed at 5-, 10-, 15-, and 20-year horizons. The behavior of these averages across time windows provides important signals about trend direction:

- Hail / Northeast & South: Averages increase as the window narrows (20yr < 15yr < 10yr < 5yr), suggesting rising risk
- Hail / Midwest: Averages also increase toward shorter windows, peaking at the 5-year average of ~$1,537
- Hurricane / Northeast: Rising trend in shorter windows (5yr = $911, vs. 20yr = $600), indicating elevated recent experience
- Hurricane / South: The 5-year average drops significantly to ~$560 from the 20-year average of ~$1,006, reflecting the absence of a major event since 2019
- Wildfire / West: Increasing trend from 20yr ($666) to 10yr ($1,172), moderating at 5yr ($1,100) — consistent with near-term elevated risk

Regional Peril Concentration

Peril	Northeast	South	Midwest	West
Hail	5.7%	22.4%	70.1%	1.8%
Hurricane	26.5%	73.5%	—	—
Wildfire	—	3.3%	1.9%	94.8%

Tail Risk Assessment from Historical Data

The relative level of tail risk across regions was assessed based on peril composition and loss volatility:

- South: Highest tail risk — exposed to both high-severity hurricanes and growing hail losses; the 2019 hurricane season produced a single-year loss exceeding $453M
- West: Second highest — wildfire losses exhibit extreme concentration in individual years (e.g., 2019: $121M, 2023: $105M, 2025: $97M)
- Northeast: Moderate-to-high tail risk — low historical frequency, but a single major hurricane event could produce catastrophic losses given dense coastal concentration
- Midwest: Lowest tail risk — high frequency hail losses are more predictable and less likely to produce catastrophically large single-year outcomes

3. CAT Model Analysis

Importance of CAT Models

Third-party stochastic CAT models are essential for pricing high-volatility perils such as hurricanes and wildfires, where historical data alone is insufficient to estimate the full distribution of potential losses. Unlike historical averages, which are limited by sample size and the under-representation of rare events, CAT models generate thousands of simulated years of events, capturing a comprehensive range of plausible loss scenarios — including scenarios not yet observed in Red Umbrella's 20-year history.

A typical CAT model consists of three interconnected modules:

- Hazard Module: Uses atmospheric science to generate stochastic event sets with event-specific conditions (wind speed, intensity) at granular geographic resolution
- Vulnerability Module: Translates physical hazard conditions into expected property damage by building class, construction type, and material
- Financial Module: Applies policy terms (limits, deductibles) to damage estimates to compute insured losses for each simulated event

The model's primary outputs for pricing purposes are: (1) Average Annual Loss (AAL), which provides a central estimate analogous to a historical average, and (2) the Exceedance Probability (EP) curve, which quantifies the full distribution of potential annual losses including tail scenarios.

AAL by Peril and Region

Peril	Northeast	South	Midwest	West	CW
Hail	$9.9M	$24.7M	$86.6M	$2.5M	$123.7M
Wildfire	—	$2.5M	$1.2M	$33.4M	$37.1M
Hurricane	$30.9M	$55.6M	—	—	$86.6M

100-Year PML by Peril and Region

Peril	Northeast	South	Midwest	West	CW
Hail	$54.2M	$62.0M	$184.2M	$6.2M	$222.7M
Wildfire	—	$15.6M	$8.2M	$201.5M	$208.7M
Hurricane	$251.7M	$399.3M	—	—	$609.1M

Ratio of 100-Year PML to AAL (Tail Risk Indicator)

The PML-to-AAL ratio is a key indicator of tail risk: a higher ratio means that a bad year is disproportionately worse than an average year, signaling the need for a higher risk load.

Peril	Northeast	South	Midwest	West	CW
Hail	5.5×	2.5×	2.1×	2.5×	1.8×
Wildfire	—	6.3×	6.6×	6.0×	5.6×
Hurricane	8.1×	7.2×	—	—	7.0×

EP Curve Insights

EP curves reveal that hurricane and hail behave very differently in the distribution of annual losses:

- Hail EP curve is relatively flat — losses at high exceedance probabilities are not dramatically lower than at low probabilities. Hail produces more consistent annual losses
- Hurricane EP curve is steep — at low exceedance probabilities (rare events), potential losses are dramatically higher relative to typical years. This steep shape is the defining characteristic of high-tail-risk perils
- The Hurricane Northeast curve is notable: despite a lower AAL than Hurricane South, the 100-year PML of $251.7M reflects the catastrophic potential of a major event striking the densely-populated Northeast coast

Potential CAT Model Errors

Two potential sources of model underestimation for hurricane risk were identified:

- Hazard Module: The model may not fully reflect increasing hurricane intensity driven by climate change, leading to underestimation of wind speeds in simulated events
- Vulnerability Module: Damage functions may underestimate losses for older commercial building stock or structures in the Northeast that have not historically been engineered to hurricane standards

4. Pure Premium Selection

Methodology

Pure premium selections were made for each peril-region combination by blending historical indications (from multiple time horizons) with model AAL indications. The appropriate blend was determined based on the following principles:

- Peril volatility: Higher-volatility perils (hurricane, wildfire) receive greater weight toward model indications, which are more stable and less sensitive to the presence or absence of major events in the observed period
- Volume of credible historical experience: Regions with more claims and less volatile experience (e.g., Midwest hail) support greater reliance on historical averages
- Trend direction: Where shorter-term averages are meaningfully higher than longer-term averages, recent experience is given additional weight; where the 5-year average appears anomalously low or high relative to the full period, it is used with caution

Pure premiums are expressed per $1 million of TIL. The model indication is derived from CAT model AAL divided by current TIL.

Hail Selections

Hail is the most stable CAT peril with relatively high frequency, making historical indications more credible than for hurricane or wildfire. Selections were anchored to the 5-year historical average, which reflects an upward trend in hail severity, blended modestly with the model indication where the two were closely aligned.

	Northeast	South	Midwest	West
Model Pure Premium	219.20	330.45	1,123.82	43.61
Hist. 20yr	155.25	366.53	1,112.65	38.04
Hist. 5yr	258.19	657.38	1,536.99	34.81
Selection	**248.44**	**575.65**	**1,536.99**	**34.81**
Blend Approach	25% model / 75% hist.	25% model / 75% hist.	100% 5yr hist.	100% 5yr hist.

Justification: For the Northeast and South, the model indication and 10-year historical average are closely aligned. The 5-year average indicates rising risk, so the selection was blended (25% model / 75% 5-year historical). For the Midwest, which has the highest hail volume and the most credible historical experience, the 5-year average was selected directly as it reflects the most current conditions. For the West, historical averages are stable and declining slightly, and the 5-year average of $34.81 is the appropriate anchor.

Hurricane Selections

Hurricane is the most volatile peril. Given the insufficiency of 20 years of experience for a peril where the full loss distribution spans return periods of hundreds of years, model indications receive the highest weight.

	Northeast	South
Model Pure Premium	684.997	743.513

Hist. 20yr	600.327	1,006.123
Hist. 10yr	876.785	1,193.238
Hist. 5yr	910.535	559.841
Selection	**732.944**	**855.944**
Blend Approach	75% model / 25% 10yr hist.	75% model / 25% 10yr hist.

Justification: For the Northeast, historical averages show increasing risk (rising from 20-year to 5-year windows), and the model indication is consistent with mid-range historical levels. A 75/25 blend toward the model was applied, using the 10-year historical average as the historical anchor to avoid over-weighting the extreme 2019 South event. For the South, the 5-year average drops dramatically to $560 (reflecting no major hurricanes since 2019), but the model at $744 and the 10-year average at $1,193 suggest long-run risk remains elevated. The 75/25 model-weighted blend produces a selection that appropriately discounts the recent quiet period while acknowledging the model's more conservative estimate relative to the extreme 20- and 15-year averages.

Wildfire Selections

Wildfire is highly volatile but slightly less so than hurricane in terms of tail risk structure. Model indications receive the primary weight, with a modest historical component to reflect recent experience trends.

	Northeast	South	Midwest	West
Model Pure Premium	—	33.05	16.05	588.72
Hist. 10yr	—	14.25	8.72	1,172.14
Hist. 5yr	—	4.63	6.39	1,100.24
Selection	**—**	**26.47**	**13.49**	**792.92**
Blend Approach	—	65% model / 35% 10yr hist.	65% model / 35% 10yr hist.	65% model / 35% 10yr hist.

Justification: For South and Midwest, historical averages decline sharply as the time window narrows, suggesting recent benign experience. However, 5-year averages are insufficient for a volatile peril. A 65/35 blend favoring the model reflects appropriate caution. For the West, the historical 10-year average of $1,172 significantly exceeds the model at $589, suggesting recent elevated experience. A 65/35 model/historical blend produces a selection of $793, which sits between the two indications and acknowledges the real possibility that recent West wildfire conditions are not fully captured in the model's baseline.

Proposed Rate Changes

Peril	Northeast	South	Midwest	West	Approach Note
Hail	+11.3%	+45.9%	+31.2%	–19.6%	Current loads understated in

					NE/South/Midwest; overstated in West
Hurricane	+9.6%	+15.5%	—	—	Model supports increases in both exposed regions
Wildfire	—	–4.0%	–1.5%	+5.9%	South/Midwest benign recent exp.; West driven by elevated historical trend

5. Risk Loads

Purpose of Risk Loads

Risk loads are premiums collected in excess of expected loss that protect an insurer against tail risk and solvency threats. They are justified by regulators when the variance of potential outcomes is sufficiently high that expected loss pricing alone would expose the insurer to ruin in a bad year. Risk loads are commensurate with the tail risk of each peril-region combination, as measured by the EP curve — particularly the PML-to-AAL ratio.

Risk Load Structure

Risk loads were set as a percentage of the selected pure premium, calibrated to the 100-year PML-to-AAL ratio. Higher ratios imply greater uncertainty relative to the mean, and therefore a larger risk load is needed to protect solvency:

	Northeast	South	Midwest	West
Hail Risk Load	15.34	16.52	33.71	2.18
Hurricane Risk Load	137.00	96.66	—	—
Wildfire Risk Load	—	3.30	1.28	88.31

Responses to Risk Load Questions

Q1: Flat risk load of $50 per million TIL for all policies

Unreasonable. Tail risk varies dramatically by peril and region. A flat load ignores the fundamental asymmetry in risk: a $50 risk load may be adequate for Midwest hail (PML/AAL ratio of ~2.1×) but would significantly underprice the hurricane risk in the South (ratio of ~7.2×) or wildfire risk in the West (ratio of ~6.0×). Risk loads must be commensurate with the actual tail exposure of each peril-region combination.

Q2: Risk loads proportional to expected CAT loss

Unreasonable. Expected loss (AAL) and tail risk are not proportional. The PML-to-AAL ratios demonstrate that high-frequency perils like hail have modest tail risk relative to their expected loss, while low-frequency catastrophic perils like hurricanes and wildfires have dramatically higher tail risk relative to expected loss. A proportional structure would over-load hail and under-load hurricane and wildfire relative to their true tail risk profiles.

Q3: Risk load exceeding expected loss + LAE for NYC metro policies

May be reasonable. For policies in dense coastal metro areas like New York City, the probability of experiencing a 100-year+ hurricane event is low, but the potential loss from such an event is extreme. The PML-to-AAL ratio for Northeast Hurricane is ~8.1×, meaning a tail event produces losses 8× the expected annual loss. In such high-concentration areas, a risk load exceeding expected loss is justifiable and consistent with actuarial standards for tail risk pricing, provided it can be supported by CAT model output.

Q4: Reinsurance covering $250M–$650M in aggregate annual CAT losses

Risk loads should decrease. Reinsurance transfers the most extreme tail scenarios to the reinsurer, effectively capping the insurer's retained loss. With a reinsurance layer covering annual losses between $250M and $650M, the insurer's financial exposure to tail events is significantly reduced. The EP curve effectively gets truncated at the attachment point for the insurer's retained layer, and risk loads should reflect only the tail risk retained by the insurer, not the full gross risk.

6. Regulatory Feedback

West Region — Wildfire Rate Denial

Regulators in the West denied Red Umbrella's wildfire rate increase (filed: +5.9%), approving only half (approved: +2.96%), citing recent adverse experience as insufficient justification for sustained elevated rates.

Recommended Strategies

- Strategy 1 — Selective Underwriting Reduction: Accept the approved rate and reduce underwriting in the highest-risk segments to maintain portfolio profitability. The segment-level loss breakdown reveals that Habitational ($49.5M in 2022) and Industrial ($18.7M in 2022) segments are disproportionately loss-prone relative to their TIL share. Red Umbrella can decline renewals or restrict new business in these segments while maintaining market presence in lower-risk categories such as Office and Other
- Strategy 2 — Regulatory Challenge with Climate Data: Submit supplementary data to regulators demonstrating that recent wildfire severity in the West reflects a structural shift rather than an anomaly. Evidence includes rising temperatures, multi-year drought cycles, increasing urban-wildland interface exposure, and third-party CAT model output showing elevated AALs that exceed the approved rate's implied expected loss
- Strategy 3 — Terms and Conditions Adjustments: Implement peril-specific wildfire deductibles or sub-limits for policies in high-exposure geographic corridors (e.g., foothills, chaparral zones). Reducing the insurer's maximum payout per event through deductibles

reduces the expected loss on the filed rate, allowing the approved premium to remain adequate at the adjusted coverage terms

- Strategy 4 — Wildfire Reinsurance: Purchase catastrophe reinsurance specifically to cover wildfire exposure in the West. By transferring a portion of tail risk to the reinsurance market, Red Umbrella reduces net retained risk and can operate profitably at the approved rate while still protecting solvency against a major event

Top Recommendation

The most effective short-term strategy is a combination of Strategies 1 and 2: tighten underwriting immediately to protect profitability by reducing exposure in the Habitational and Industrial segments (which represent the highest loss-to-TIL concentration), while simultaneously engaging regulators with climate data and model support to establish the case for full rate approval in the next filing cycle. Strategy 3 (deductibles) provides additional protection at the policy level. This dual approach manages short-term risk while building a long-term solution.

Northeast Region — Hurricane Rate Scrutiny

Regulators in the Northeast noted that Red Umbrella's filed CAT pure premiums are significantly higher in the Northeast than in other regions relative to recent historical losses, and requested additional support before approving the filing.

Draft Regulatory Response

To: State Regulatory Authority

Re: Red Umbrella Insurance — Northeast CAT Rate Filing

Thank you for your review of Red Umbrella's catastrophe rate filings. We appreciate the opportunity to provide additional context regarding our Northeast region CAT pure premiums, which may appear elevated relative to recent loss experience.

While historical losses in the Northeast have been modest over the past 20 years, this observation reflects the low-frequency, high-severity nature of Northeast hurricane risk — not an absence of risk. The Northeast faces significant tail risk from a rare but potentially devastating hurricane event that cannot be estimated using historical averages alone.

Our filed rates are supported by industry-standard third-party catastrophe model output. Key metrics include:

- 100-year PML for Northeast Hurricane: $251.7 million
- AAL for Northeast Hurricane: $30.9 million
- Ratio of 100-year PML to AAL: 8.1×, indicating extreme tail risk relative to expected loss

This ratio demonstrates that while average annual losses are modest, a single event could exceed $250 million — threatening the insurer's ability to meet its obligations to policyholders. The risk load included in our rates is commensurate with this tail exposure and is consistent with regulatory allowances for high-variance CAT perils. Historical data alone is insufficient for pricing rare, high-severity events, which is precisely the use case for which catastrophe models were developed.

We welcome further discussion and are happy to provide the full EP curve output and sensitivity analyses supporting our filing.

Respectfully, Red Umbrella Insurance Actuarial Pricing Team

7. Conclusions

This analysis produced CAT pure premium selections for all relevant peril-region combinations by synthesizing 20 years of adjusted historical loss data with third-party stochastic CAT model output. Several overarching conclusions emerged:

- Hurricane is the highest-tail-risk peril in Red Umbrella's portfolio, with a 100-year countrywide PML of $609M — approximately 7× the model AAL. Rate increases are warranted in both the Northeast and South
- Wildfire in the West represents the most dynamically evolving risk. Recent historical experience significantly exceeds the model AAL, suggesting the model may underestimate current-environment risk. A blend weighted toward the model but incorporating recent historical trends was used
- Hail is the most credibly priced by historical data due to its relative frequency and lower tail risk. Rate adjustments reflect upward trends in shorter-term averages for most regions
- Risk loads should be differentiated by peril and region, and calibrated to tail risk as measured by the PML-to-AAL ratio. Flat or proportional-to-AAL risk load structures are inappropriate for a portfolio with this degree of peril heterogeneity
- In regulatory interactions, EP curve metrics — particularly the PML-to-AAL ratio — provide the most compelling quantitative support for filed CAT rates, especially for low-frequency, high-severity perils like Northeast hurricane

The full analysis, supporting calculations, and model output are documented in the accompanying Excel workbook.